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1. Introduction

In this lab, we implemented a conditional denoising diffusion probabilistic model (DDPM) for the i-CLEVR dataset. The goal is to generate synthetic images according to given multi-label object conditions, where each condition may contain one to multiple objects with different colors and shapes.

To evaluate the generation quality in an objective way, the generated images are fed into a pretrained ResNet18-based evaluator provided by the TA. The final performance is measured by the classification accuracy on test.json and new_test.json.

In our implementation, we first built a baseline conditional DDPM using multi-hot label encoding, a conditional U-Net, cosine noise schedule, and standard DDPM noise prediction loss. Then, to further improve conditional controllability, we introduced classifier-free guidance (CFG), which significantly improved the final evaluation accuracy.

2. Implementation Details

2.1 Dataset and Label Encoding

The i-CLEVR dataset contains 24 object categories formed by 8 colors and 3 shapes. objects.json provides the mapping from object name to class index.

For training, train.json is a dictionary whose keys are image filenames and whose values are lists of object names in the image. Each image was resized to 64×64, converted to RGB, transformed into tensor format, and normalized to $[-1,1]$using:

$$\text{Normalize}((0.5,0.5,0.5),(0.5,0.5,0.5)).$$

Since one image may contain multiple objects, we used a **24-dimensional multi-hot vector** to represent the conditional labels. For example, if an image contains yellow sphere and gray cylinder, the corresponding entries of these two categories are set to 1 and all others remain 0.

2.2 Conditional DDPM

We implemented a conditional DDPM to generate images according to the given object conditions. During training, a clean image $x_0$is corrupted into a noisy image $x_t$by the forward diffusion process:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \epsilon \sim \mathcal{N}(0, I).$$

The model is trained to predict the added Gaussian noise $\epsilon$from the noisy image x_t, timestep t, and condition label c. The simplified DDPM loss is:

$$L = \mathbb{E}_{x_0,t,\epsilon}[\| \epsilon - \epsilon_\theta(x_t, t, c) \|^2].$$

2.3 Noise Schedule

Instead of using a linear noise schedule, we adopted a **cosine noise schedule**, because it provides a smoother change of signal-to-noise ratio during the diffusion process. Compared with a simple linear schedule, the cosine schedule preserves image information better in the early timesteps and is generally easier to learn in the reverse denoising process. Compared with a simple linear schedule, the cosine schedule preserves image information better in the early timesteps and is generally easier to learn in the reverse denoising process.

2.4 Time Embedding and Condition Embedding

To make the model aware of the current diffusion step, we used sinusoidal time embedding. The timestep $t$is first encoded into a vector and then projected through MLP layers.

For condition embedding, the 24-dimensional multi-hot label vector is projected by fully connected layers into the same embedding dimension. The final conditioning vector is obtained by:

$$e = e_{\text{time}} + e_{\text{label}}.$$

This combined embedding is injected into each residual block of the U-Net so that the model can perform denoising conditioned on both timestep and object labels.

2.5 Conditional U-Net Architecture

The noise prediction network is a conditional U-Net. The input is the noisy image $x_t \in \mathbb{R}^{3\times64\times64}$, together with the timestep embedding and multi-label condition embedding.

The U-Net consists of:

- an initial convolution layer,
- several downsampling residual blocks,
- middle residual blocks,
- several upsampling residual blocks with skip connections,
- a final convolution layer to predict the noise.

The output has the same spatial shape as the input image and represents the predicted noise.

2.6 Classifier-Free Guidance (CFG)

To improve the model's adherence to the input conditions, we further introduced **classifier-free guidance (CFG)**.

During training, with probability 0.1, the conditional label vector is replaced by an all-zero vector. This enables the model to learn both conditional and unconditional denoising.

During sampling, both conditional and unconditional predictions are computed:

$$\epsilon_{\text{guided}} = \epsilon_{\text{uncond}} + s \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}}),$$

where $s$is the guidance scale.

We experimented with different guidance scales and found that **guidance scale = 3.5** gave the best final accuracy when using checkpoint epoch_190.pth.

3. Results and Discussion

3.1 Baseline Result

The baseline conditional DDPM without classifier-free guidance achieved:

- test.json accuracy = 0.736111
- new_test.json accuracy = 0.761905

This shows that the model was already able to generate visually recognizable objects and follow the given object conditions to some extent.

3.2 Effect of Classifier-Free Guidance

After adding classifier-free guidance and continuing training, the generation quality and evaluation accuracy improved significantly.

We tested different guidance scales. The best result was obtained at:

- checkpoint: epoch_190.pth
- guidance scale: 3.5

The final accuracy was:

- test.json accuracy = 0.805556
- new_test.json accuracy = 0.904762

This is a clear improvement over the baseline model, especially on new_test.json. The result indicates that classifier-free guidance effectively strengthens the conditional control of the diffusion model and helps the generated images better match the target label sets. The final selected result was produced from epoch_190.pth with guidance scale = 3.5, yielding 0.805556 on test.json and 0.904762 on new_test.json.

3.3 Denoising Process

We also visualized the denoising process for the condition:

["red sphere", "cyan cylinder", "cyan cube"].

Starting from pure Gaussian noise, the generated image gradually becomes clearer as the reverse diffusion process proceeds. The intermediate results show that the model first recovers the coarse layout and then progressively refines the object shapes and colors.

3.4 Discussion

The experimental results show that the model can successfully generate multi-object images conditioned on different object combinations. The generated images are visually clear, and most of them contain the intended colors and shapes.

Compared with the baseline model, classifier-free guidance improved the conditional controllability and final evaluator accuracy. However, the

performance still fluctuates depending on the sampling randomness and the selected guidance scale. When the guidance scale is too small, the generated images may not follow the condition strongly enough. When it is too large, the sampling may become less stable.

In our experiments, a guidance scale of 3.5 provided the best overall trade-off.

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4. Conclusion

In this lab, we successfully implemented a conditional DDPM for the i-CLEVR dataset. The model takes multi-label conditions as input and generates corresponding images through a reverse denoising process.

By adopting a cosine noise schedule, conditional U-Net, and multi-hot label embedding, the baseline model already achieved reasonable performance. Furthermore, classifier-free guidance significantly improved the final results.

Our best model, obtained from checkpoint epoch_190.pth with guidance scale 3.5, achieved:

- test.json accuracy = 0.805556
- new_test.json accuracy = 0.904762

These results demonstrate that diffusion-based conditional image generation can effectively model the i-CLEVR dataset and produce images consistent with complex multi-object conditions. The final selected model was epoch_190.pth with guidance scale = 3.5, achieving 0.805556 on test.json and 0.904762 on new_test.json.